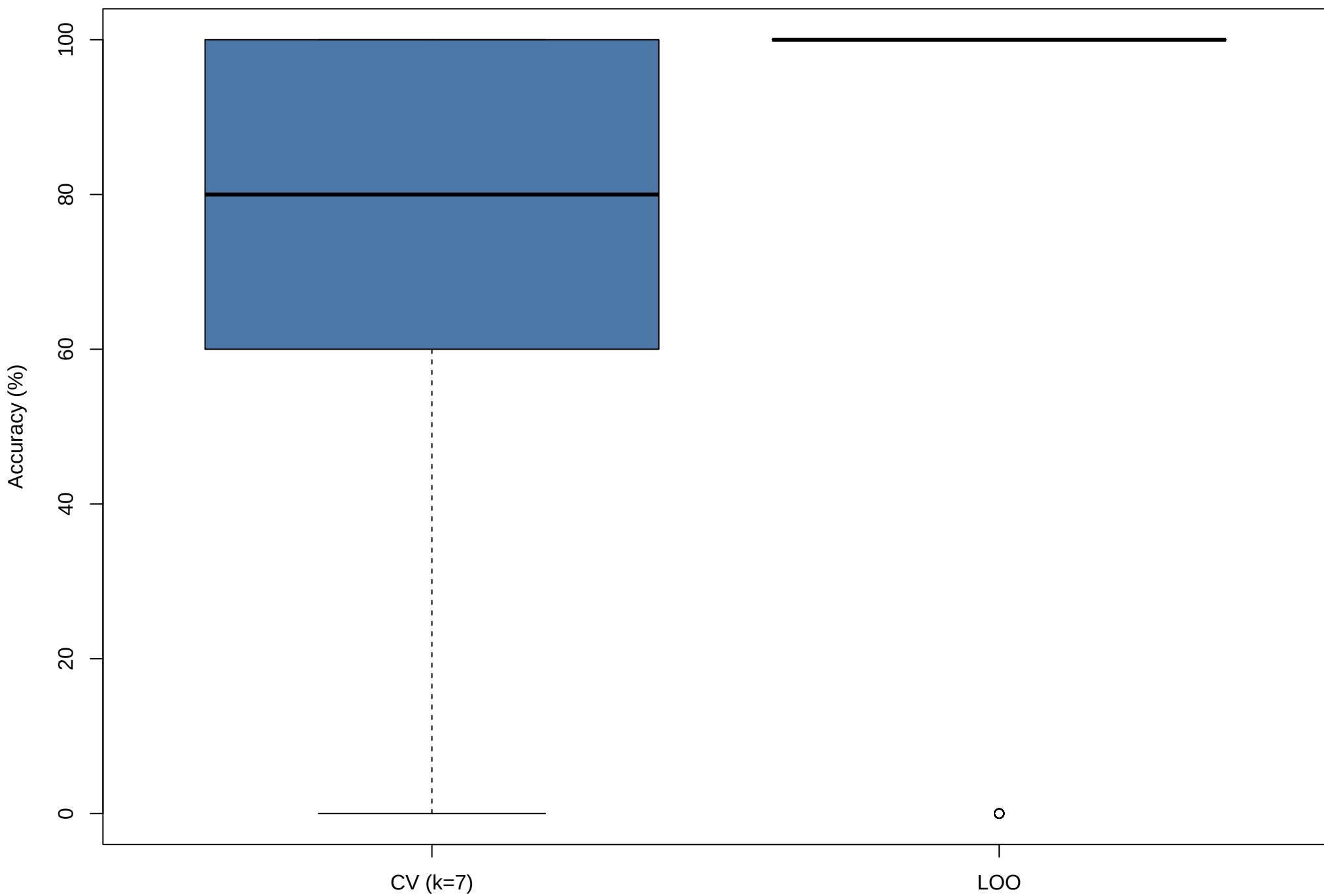
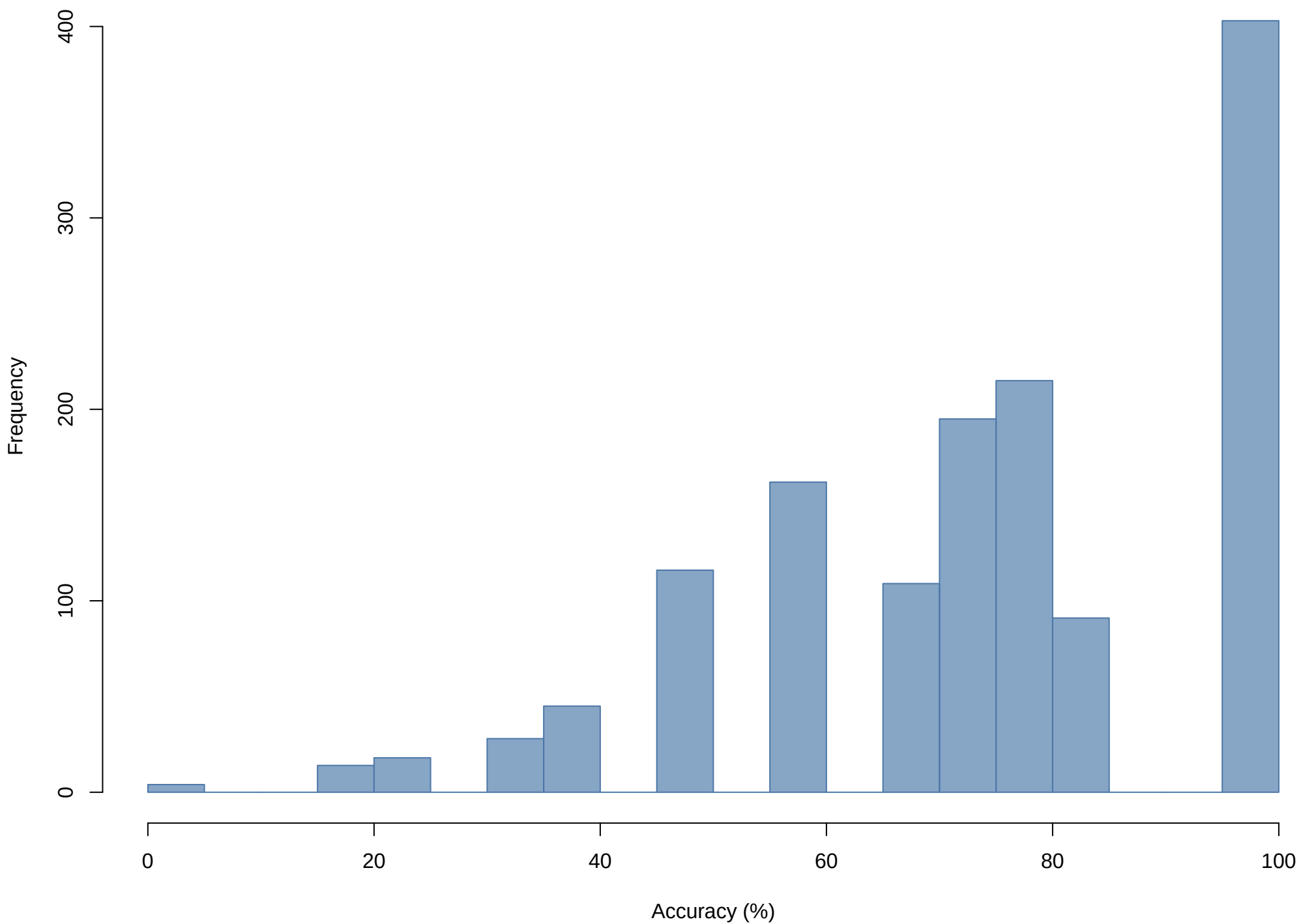


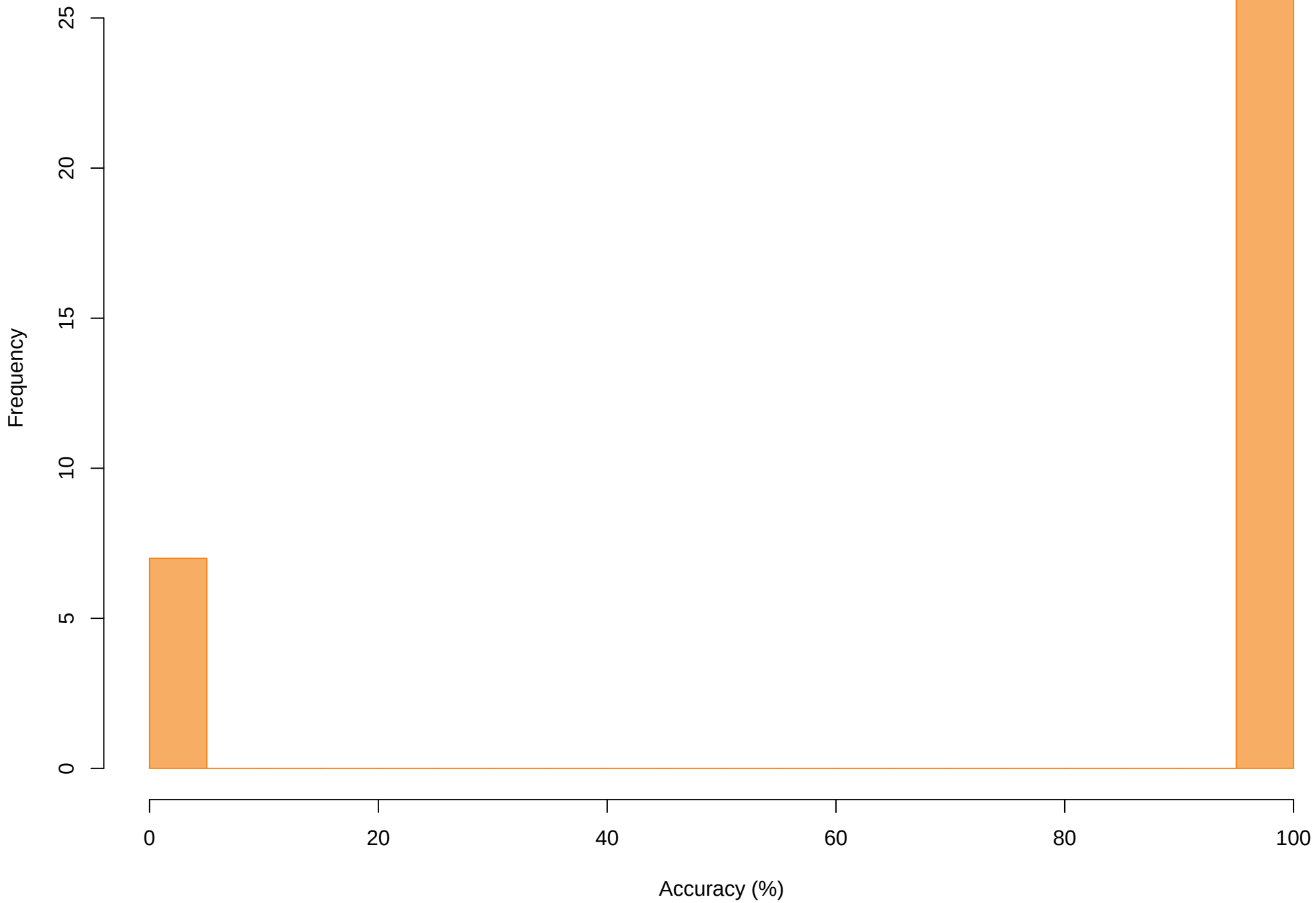
Cross-validation accuracy (Kennard–Stone train pool)



CV (k=7) accuracy histogram



LOO accuracy histogram

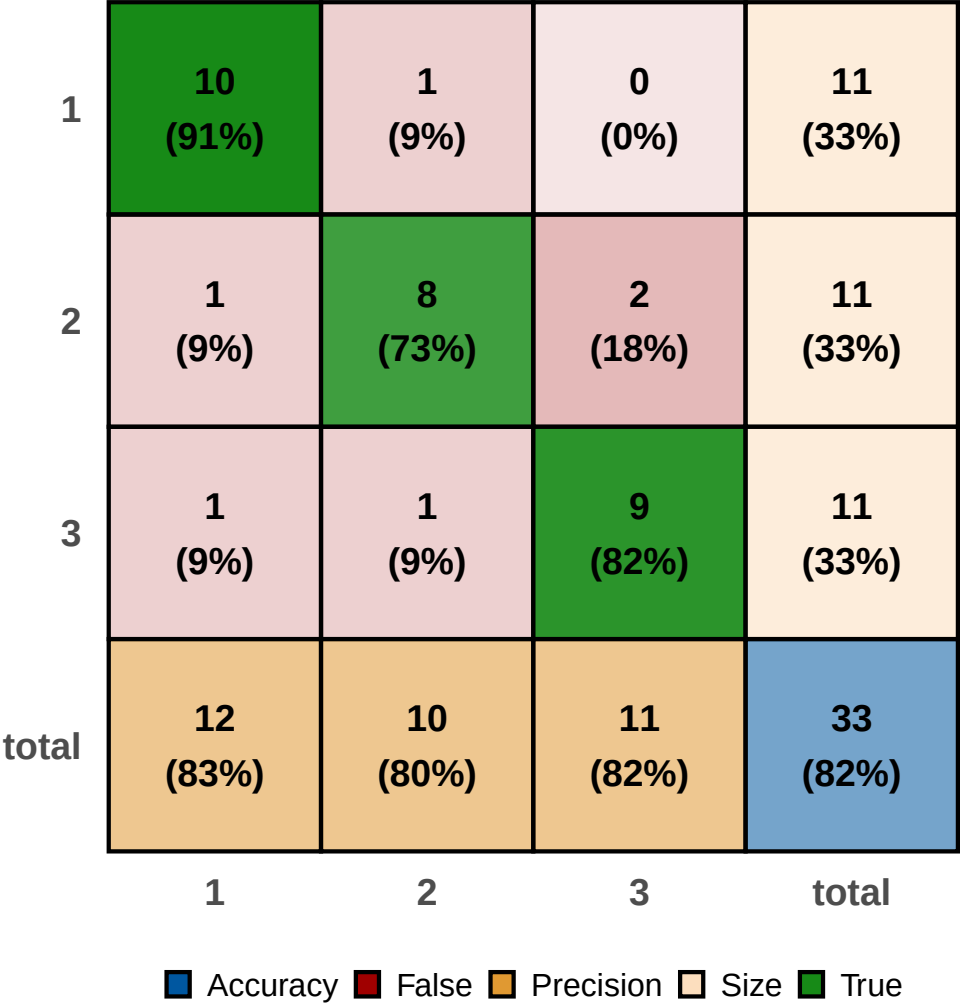


Train confusion matrix

actual \ pred	1	2	3
	1	2	3
1	10	1	0
2	1	8	2
3	1	1	9

Train confusion table

Train Set (Kennard–Stone pool; model report)

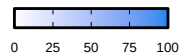


3. 3rd Place

Train Set (Kennard–Stone pool; model report)

Probability Heatmap

	1	2	3	Exp
2V	100	0	0	1
2R	17	73	10	1
2S	83	17	0	1
2U	90	10	0	1
2A	67	32	1	1
2D	99	1	0	1
2L	63	36	1	1
2O	99	1	0	1
2N	77	23	1	1
2K	100	0	0	1
2Q	96	4	0	1
2AQ	23	70	7	2
2AC	25	69	6	2
2AJ	1	34	65	2
2AL	52	46	2	2
2AA	1	32	67	2
2AM	7	71	22	2
2AI	4	64	32	2
2AG	22	71	7	2
2AN	14	74	12	2
2Y	7	70	23	2
2AP	31	65	5	2
2AS	2	48	50	3
2AT	1	27	72	3
2AU	53	45	2	3
2AV	1	24	75	3
2AW	2	51	46	3
2AX	0	4	96	3
2AY	1	20	79	3
2AZ	0	2	98	3
2BA	0	2	98	3
2BB	0	5	95	3
2BC	0	0	100	3



Holdout confusion matrix

actual \ pred	1	2	3
	1	2	3
1	11	0	0
2	0	10	1
3	0	0	0

Holdout confusion table

Holdout (Kennard–Stone; model report)

	actual \ pred			total
	1	2	3	
1	11 (100%)	0 (0%)	0 (0%)	11 (50%)
2	0 (0%)	10 (91%)	1 (9%)	11 (50%)
3	0 (0%)	0 (0%)	0 (0%)	0 (0%)
total	11 (100%)	10 (100%)	1 (0%)	22 (95%)
	1	2	3	total

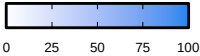
■ Accuracy ■ False ■ Precision ■ Size ■ True

3. 3rd Place

Holdout (Kennard–Stone; model report)

Probability Heatmap

	1	2	3	Exp
2B	78	21	1	1
2C	78	22	1	1
2E	77	22	1	1
2F	73	26	1	1
2G	90	10	0	1
2H	94	6	0	1
2I	81	19	1	1
2J	74	26	1	1
2M	78	22	1	1
2P	97	3	0	1
2T	100	0	0	1
2AB	5	65	29	2
2AD	43	54	3	2
2AE	9	73	18	2
2AF	31	64	5	2
2AH	16	74	10	2
2AK	5	67	28	2
2AO	24	70	7	2
2AR	1	26	74	2
2W	42	55	3	2
2X	29	66	5	2
2Z	3	52	45	2



External validation — class probabilities and predictions (accuracy = 100.00%)

substrate	actual_class	predicted_class		
		1	2	3
2BD	2	22.81	70.23	6.96
2BE	2	2.70	52.99	44.31
2BF	1	52.10	45.91	1.99
2BG	2	3.76	60.13	36.11
2BH	3	0.01	0.28	99.72
2BI	3	0.61	21.14	78.25

External confusion matrix

	pred			
actual	1	2	3	
1	1	0	0	
2	0	3	0	
3	0	0	2	

External confusion table

External Validation (model report)

1	1 (100%)	0 (0%)	0 (0%)	1 (17%)
2	0 (0%)	3 (100%)	0 (0%)	3 (50%)
3	0 (0%)	0 (0%)	2 (100%)	2 (33%)
total	1 (100%)	3 (100%)	2 (100%)	6 (100%)
	1	2	3	total

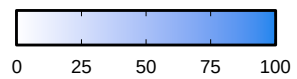
■ Accuracy ■ False ■ Precision ■ Size ■ True

1. 1st Place

External Validation (model report)

Probability Heatmap

	1	2	3	Exp
2BD	23	70	7	2
2BE	3	53	44	2
2BF	52	46	2	1
2BG	4	60	36	2
2BH	0	0	100	3
2BI	1	21	78	3



1. 1st Place